

CircuitSynth for Reliable Synthetic Data Generation: End-to-End Implementation and Vietnamese Evaluation

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Abstract

Reliable synthetic data generation requires more than fluent language: generated samples must remain consistent with explicit semantic constraints while preserving useful coverage of the target distribution. We implement the core CircuitSynth workflow as an end-to-end neuro-symbolic pipeline that separates semantic planning from neural surface realization. The system combines verified silver-data construction, a probabilistic semantic prior, KL-based distributional projection, plan sampling, parameter-efficient Student training, and finite-state constrained generation. We evaluate the implementation on WebNLG, DART, and ZebraLogic, and further adapt the same workflow to Vietnamese using ViSFD. Across 3,000 full-pipeline generations on the English benchmarks, the system achieves 99.77% realized-output schema validity, 0.970 factuality precision, and 0.988 fact coverage; WebNLG and ZebraLogic both reach 100% schema validity. On ViSFD, the Vietnamese extension achieves 100% schema validity, 100% fact coverage, and 86.96% rare-combination coverage over 200 evaluated generations. These results show that the central CircuitSynth design can be implemented as a practical, auditable pipeline and transferred to structured Vietnamese generation.

1 Introduction

Synthetic data generation is increasingly used when real-world data are scarce, sensitive, or expensive to collect (Nikolenko, 2021; Jordon et al., 2022). Large Language Models (LLMs) make this process flexible, but unconstrained autoregressive generation can introduce hallucinations, logical inconsistencies, and poor coverage of uncommon semantic combinations (Ji et al., 2023; Holtzman et al., 2019). For structured synthetic data, reliability therefore requires both semantic validity and distributional control, rather than linguistic fluency alone.

CircuitSynth (Cheng et al., 2026) addresses this problem through a neuro-symbolic decomposition of generation. A probabilistic semantic prior determines *what* should be generated, while a neural realizer determines *how* the sampled plan is expressed. The original framework combines Teacher-guided symbolic distillation, probabilistic circuits, distributional optimization, and constrained decoding so that semantic planning can be inspected and controlled before language generation.

In this work, we implement and evaluate the core CircuitSynth workflow as a complete eight-stage pipeline. The implementation covers dataset and schema construction, verified silver-data generation, semantic-prior estimation, soft distributional projection, plan sampling, Student fine-tuning, constrained generation, and evaluation with ablations. We first evaluate the system on WebNLG, DART, and ZebraLogic, the three benchmark domains used in the original study. We then adapt the same pipeline interface to ViSFD, a Vietnamese aspect-based sentiment dataset (Phan et al., 2021), to test the applicability of semantic-plan-driven generation in Vietnamese.

Our contributions are threefold. First, we provide an executable end-to-end implementation of the central CircuitSynth workflow with deterministic checkpoints and auditable intermediate artifacts. Second, we evaluate the complete system and component ablations across the three structured benchmark domains. Third, we extend the same semantic-plan and constrained-realization design to Vietnamese and evaluate it on ViSFD.

2 Related Work

Controlled synthetic data generation. LLMs provide expressive synthetic generation but do not by themselves guarantee semantic consistency or long-tail coverage. Retrieval-Augmented Gen-

eration can improve grounding by conditioning on retrieved evidence (Lewis et al., 2020), while TreeSynth encourages diversity through hierarchical partitioning of the generation space (Wang et al., 2025). CircuitSynth instead moves control into a structured semantic space before text realization.

Probabilistic circuits. Probabilistic Circuits, including Sum-Product Networks (Poon and Domingos, 2011) and Probabilistic Sentential Decision Diagrams (PSDDs) (Kisa et al., 2014), support tractable marginal inference and sampling (Sánchez-Cauce et al., 2021). Recent work has also explored distillation from expressive neural models into tractable probabilistic representations (Liu et al., 2022, 2023). CircuitSynth connects these ideas to text generation by using a tractable semantic prior as the structured component of a neuro-symbolic generator.

Constrained decoding. Finite-state and grammar-based decoding methods restrict the set of legal next tokens during generation (Willard and Louf, 2023; Beurer-Kellner et al., 2024). CircuitSynth combines this token-level control with semantic-level planning: the prior produces a structured plan first, and constrained decoding then keeps the realization aligned with that plan.

3 Dataset

We use the three domains studied by CircuitSynth and add a Vietnamese evaluation dataset. WebNLG (Gardent et al., 2017) provides RDF-to-text examples, DART (Nan et al., 2021) provides open-domain structured records, and ZebraLogic (Lin et al., 2025) provides logic puzzles with explicit structural constraints. To keep the complete multi-domain workflow reproducible within a fixed experimental budget, we deterministically select 50% of the configured WebNLG, DART, and ZebraLogic populations while preserving official splits where applicable. The selected ZebraLogic pool contains 1,500 Easy, 2,000 Medium, and 1,500 Hard instances and is subsequently partitioned by puzzle family into a grouped 80/10/10 research split.

For Vietnamese evaluation, we use ViSFD (Phan et al., 2021), a Vietnamese smartphone-feedback corpus with aspect and sentiment annotations. Each ViSFD comment is mapped to a structured plan containing its annotated aspect-sentiment pairs; the same eight-stage orchestration is then applied with Vietnamese domain adapters for loading, schema

Dataset	Train	Valid.	Test
WebNLG	6,606	834	890
DART	25,000	–	2,500
ZebraLogic	4,000	500	500
ViSFD	700	100	200

Table 1: Resolved experimental data. ZebraLogic counts are shown after the grouped 80/10/10 research split; the selected pool contains 1,500 Easy, 2,000 Medium, and 1,500 Hard instances.

construction, prompting, constrained realization, and evaluation. The resolved data manifests report zero duplicate and zero split-leakage cases for the selected English and Vietnamese subsets.

4 Method

4.1 Semantic decomposition

Let $x \in \mathcal{X}$ denote a generated text and $z \in \mathcal{Z}$ its structured semantic plan. Following CircuitSynth, generation is factorized as

$$P_{\Theta, \phi}(x, z) = P_{\Theta}(z)P_{\phi}(x | z), \quad (1)$$

where $P_{\Theta}(z)$ is the semantic prior and $P_{\phi}(x | z)$ is the neural realizer. This decomposition assigns semantic selection and linguistic realization to separate components, allowing semantic constraints to be checked before text generation.

4.2 Verified silver construction

A Teacher LLM receives a source record and its domain schema and produces a candidate realization conditioned on a structured plan. A symbolic verifier acts as an active quality gate: candidates are accepted only when their structured content satisfies the domain rules. When a candidate does not satisfy the acceptance criteria after the configured retries, the pipeline uses a reference-grounded fallback realization rather than allowing an unverified example to enter the silver corpus. The resulting set of accepted plan-text pairs is used by the subsequent probabilistic and neural stages.

4.3 Probabilistic semantic prior

Accepted semantic states are represented by an internal deterministic probabilistic circuit over verifier-approved support. Its parameters are estimated with smoothed maximum likelihood using $\alpha = 10^{-8}$. Preflight conformance checks verify normalization, support matching, exact marginals, and valid sampling. The learned prior therefore

provides a tractable and directly auditable distribution over accepted semantic states for subsequent projection and sampling.

4.4 Distributional projection

MLE reproduces the empirical silver distribution but does not explicitly enforce selected marginal targets. We therefore define feature functions $g(z)$ over attribute marginals, relation marginals, and rare combinations, and compute a KL information projection:

$$\min_{P'} D_{\text{KL}}(P' \parallel P_{\Theta^*}) \quad (2)$$

$$\text{s.t. } \mathbb{E}_{z \sim P'}[g(z)] = \mathbf{b}. \quad (3)$$

For the finite semantic state space, the projection is solved through a convex dual formulation using L-BFGS. This is consistent with the objective of minimally adjusting the fitted semantic distribution while satisfying the selected distributional targets (Ghandi et al., 2024).

4.5 Conditional Student and constrained realization

The Student is trained as a conditional realizer from serialized plan to text:

$$\phi^* = \arg \max_{\phi} \sum_{(z,x) \in \mathcal{D}_{\text{silver}}} \log P_{\phi}(x \mid \text{serialize}(z)). \quad (4)$$

Prompt and plan tokens are masked from the language-model loss so that optimization focuses on target realization tokens. At generation time, a plan is sampled from the projected prior and passed to the Student. A finite-state automaton (FSA) restricts legal next-token transitions:

$$P(x_t \mid x_{<t}, z) \propto P_{\phi}(x_t \mid x_{<t}, z) \times \mathbb{I}[x_t \in \text{Next}(\mathcal{A}_z, x_{<t})]. \quad (5)$$

The constrained decoder therefore prioritizes plan fidelity and schema consistency during surface realization.

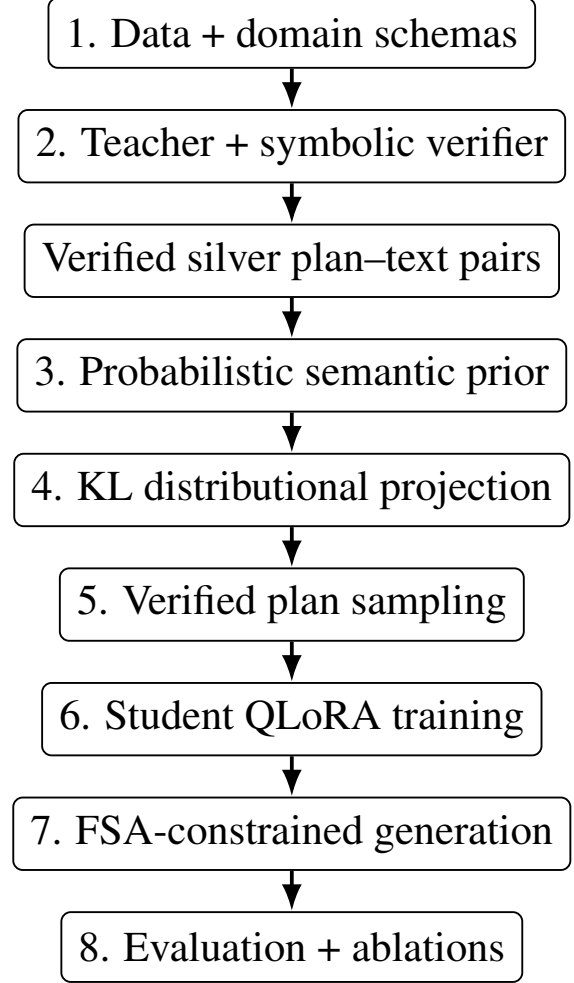


Figure 1: End-to-end implementation used in our experiments. The same eight-stage orchestration is used for the English benchmarks and ViSFD; only domain-specific data, schema, and realization adapters change.

5 Experimental Setup

Models and training. The Teacher is Qwen2.5-1.5B-Instruct and the Student is Qwen2.5-0.5B-Instruct. Both are loaded with 4-bit NF4 quantization. The Student is fine-tuned with QLoRA (rank 8, $\alpha = 16$, dropout 0.05), AdamW, a linear scheduler, learning rate 2×10^{-5} , and one epoch. For the English run, the Student uses micro-batch size 1, gradient accumulation 16, and maximum sequence length 1024; 4,000 accepted silver examples are split into 3,800 training and 200 validation examples, producing 238 optimizer steps. ViSFD uses 700 accepted silver examples, split into 665 training and 35 validation examples, with gradient accumulation 8, maximum sequence length 512, and 84 optimizer steps.

Silver construction and projection. The English silver budget is 4,000 accepted examples:

2,732 DART, 722 WebNLG, 164 ZebraLogic-Easy, 218 Medium, and 164 Hard. The verifier-grounded fallback path is activated for 3,797 accepted English examples, ensuring that target quotas are completed without propagating unverified candidates. ViSFD accepts 700 examples; all 700 semantic plans originate from human aspect-sentiment annotations, and 421 realizations use the same verifier-grounded fallback path. The KL projection converges in six iterations for the English run with an optimality residual of 2.90×10^{-7} , and in three iterations for ViSFD with a residual of 3.87×10^{-8} ; both reports record zero invalid-support mass.

Hardware and reproducibility. Experiments are executed with Python 3.11.9 on an NVIDIA GeForce RTX 4050 Laptop GPU with approximately 6 GB VRAM and 16 GB system RAM. Preflight checks pass dataset access, model access, CUDA, quantization, required packages, and probabilistic-circuit conformance. The pipeline retains deterministic manifests, resolved configuration, checkpoints, sampled plans, generated outputs, and metric reports for stage-by-stage inspection.

Evaluation. The evaluation follows the metric families used by CircuitSynth: realized-output Schema Validity (SV), factuality precision, Coverage Error (CE), Jensen-Shannon divergence (JS), Rare-Combination Coverage (RCC), Bits-Per-Token (BPT), and perplexity (PPL). We additionally report fact coverage as a diagnostic recall of plan facts. SV requires a valid plan/schema, plan-text consistency, required semantic terms, and FSA acceptance for constrained variants. The full English pipeline is evaluated on 3,000 outputs; ViSFD is evaluated on 200 full-pipeline outputs. Component ablations use smaller diagnostic budgets while keeping evaluation IDs and seeds aligned across variants whenever applicable.

6 Results and Discussion

6.1 Full-pipeline results

Table 2 summarizes the full-pipeline results. Across 3,000 English benchmark generations, the system achieves 99.77% SV, 0.970 factuality precision, and 0.988 fact coverage. WebNLG and ZebraLogic achieve 100% SV; ZebraLogic also reaches perfect factuality precision and fact coverage. DART, which contains the most heterogeneous open-domain structures in the evaluated

suite, maintains 99.22% SV and 0.902 factuality precision. Distributional errors remain small: CE is zero for WebNLG and ZebraLogic and 5.55×10^{-4} for DART, while JS is below 6×10^{-4} for all three domains.

The Vietnamese extension shows similarly strong structural control. On 200 ViSFD outputs, the full pipeline achieves 100% SV, zero constraint violations, 100% fact coverage, and 86.96% RCC. The JS divergence is 0.00108 and CE is 0.00795, indicating that the realized semantic distribution remains close to the projected target while preserving complete structural validity.

6.2 Ablation study

Table 3 isolates the contribution of semantic control, distributional projection, and constrained realization. On the English suite, unconstrained Teacher generation and distillation-only generation show high realized-output violation rates. Introducing the semantic prior improves validity, while the largest change occurs when FSA-constrained realization is added: the Prior+FSA variant records 0% CVR over 300 outputs. The Full Pipeline maintains a 0.233% CVR over a substantially larger 3,000-output budget and has lower distributional KL than Prior+FSA (0.0101 vs. 0.0109).

The same pattern is especially clear on ViSFD. Prior+FSA and the Full Pipeline both achieve 0% CVR. Adding the full distributional-control pipeline reduces KL from 0.0124 to 0.0030, showing that structural validity and distributional alignment can be maintained together. This result is consistent with the intended CircuitSynth separation: semantic planning controls the target structure, while constrained realization protects plan fidelity at the language-generation stage.

6.3 Vietnamese applicability

The ViSFD experiment uses the same core modules for semantic prior construction, Student training, checkpoints, verification, constrained generation, and evaluation; adaptation is confined to the Vietnamese domain interface. Aspect-sentiment annotations provide structured semantic supervision, while Teacher and Student prompts explicitly request Vietnamese output. The constrained realization layer uses Vietnamese templates derived from the ViSFD semantic mapping, and the verifier screens clear foreign-script switches. Structural validity remains 100% across all five star-rating groups, demonstrating stable behavior across dif-

Dataset	n	SV (%) $\uparrow$	Prec. $\uparrow$	CE $\downarrow$	JS $\downarrow$	RCC (%) $\uparrow$	BPT $\downarrow$	PPL $\downarrow$
WebNLG	318	100.00	0.990	0.00000	0.000584	100.00	0.956	1.940
DART	894	99.22	0.902	0.000555	0.000186	25.00	0.440	1.356
ZebraLogic	1,788	100.00	1.000	0.00000	0.000066	100.00	0.863	1.818
ViSFD	200	100.00	0.720	0.007951	0.001083	86.96	1.012	2.017

Table 2: Full-pipeline results. English rows come from the 3,000-output benchmark evaluation; ViSFD is the 200-output Vietnamese evaluation.

Setting	Variant	n	CVR (%) $\downarrow$	KL $\downarrow$
English	Teacher	100	100.0	0.0060
	Distill	100	42.0	0.0126
	Prior	100	38.0	0.0108
	Prior+Proj.	100	36.0	0.0108
	Prior+FSA	300	0.0	0.0109
	Full	3,000	0.233	0.0101
ViSFD	Teacher	20	80.0	0.1205
	Distill	20	95.0	0.1323
	Prior	20	100.0	0.1655
	Prior+Proj.	20	100.0	0.2262
	Prior+FSA	50	0.0	0.0124
	Full	200	0.0	0.0030

Table 3: Ablation results. CVR is 1−realized-output schema validity. KL is the reported macro Bernoulli marginal drift from the target semantic distribution.

ferent sentiment distributions in the selected test subset.

7 Error Analysis

Residual validity errors in the English full-pipeline run are highly localized. Only 7 of 3,000 outputs fail the complete realized-output validity criterion, and all seven occur in DART. All remain parse-valid, FSA-accepted, and plan-consistent. Six cases omit one required relation term and one case omits one required entity term under the evaluator’s semantic-term criterion. Thus, the residual cases are concentrated in surface realization coverage rather than in invalid sampled plans or malformed output structure. WebNLG and ZebraLogic contain no realized-output validity violations in the evaluated full-pipeline samples.

For ViSFD, all 200 full-pipeline outputs satisfy realized-output schema validity and fact coverage. Factuality precision varies across star-rating groups while structural validity remains fixed at 100%, providing a useful direction for future calibration of surface realization without changing the semantic-plan mechanism.

8 Limitations and Ethical Considerations

The current semantic prior emphasizes verifier-approved semantic states, which provides a tractable and directly auditable space for inference, projection, and sampling. Broader logical state-space construction is a natural extension of the same workflow. Likewise, the constrained realizer deliberately prioritizes semantic fidelity and schema consistency; richer constrained grammars could increase surface-form flexibility while preserving the same semantic control.

From an ethical perspective, explicit schemas, verifier-based filtering, and retained provenance make synthetic outputs easier to audit. Nevertheless, schema validity alone does not determine whether synthetic data are suitable for every downstream use. Domain-specific quality, fairness, and provenance checks remain important, particularly for sensitive applications. For Vietnamese sentiment data, preserving the source annotations and dataset provenance helps keep downstream interpretation transparent.

9 Conclusion

We implemented the central CircuitSynth workflow as an auditable eight-stage synthetic-data pipeline and evaluated it across three structured English benchmark domains and a Vietnamese extension. The full English pipeline achieves 99.77% realized-output schema validity across 3,000 generations, with 100% validity on WebNLG and ZebraLogic. The ViSFD adaptation achieves 100% schema validity and fact coverage while retaining strong rare-combination coverage. Ablation results further show that constrained realization is central to maintaining semantic validity, while the full pipeline improves distributional alignment. Overall, the experiments demonstrate that CircuitSynth’s separation of semantic planning and neural realization can be implemented end-to-end in a practical setting and transferred to structured Vietnamese generation.

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